

Omar Ayman Bakr

Fresh Machine Learning Engineer

✉ Email 📞 +20 1012772518 📍 Cairo, Egypt
🌐 Portfolio 🔗 LinkedIn 🐙 Github 🏆 LeetCode

PROFESSIONAL SUMMARY

A recent Mechatronics Engineering graduate specializing in Machine Learning with a focus on Computer Vision, backed by a rigorous foundation in Linear Algebra, Probability, and advanced Data Structures. Demonstrates strong problem-solving ability with over [200+ LeetCode problems solved](#). Seeking to leverage a strong quantitative background and vision-focused expertise to develop innovative solutions in Computer Vision and Deep Learning applications.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, C, MATLAB, SQL
- **AI Frameworks & Libraries:** PyTorch, Scikit-learn, LangChain, Pandas, NumPy, imblearn, Matplotlib, Seaborn, Plotly
- **Backend & APIs:** FastAPI, Uvicorn, Pydantic, REST API design, async I/O, Docker Compose
- **Developer Tools:** Git, GitHub, VS Code, PyCharm, Jupyter Notebook, Conda, uv, Linux/Fedora, Docker
- **Databases:** PostgreSQL, pgAdmin4, MongoDB (Motor)
- **Core Concepts:** Data Structures, Algorithms, Calculus (Single & Multivariable), Linear Algebra, Probability

PROJECTS

NotebookLLM-minus — Retrieval-Augmented Document Q&A Service

In progress — August 2026 to present

- Building a NotebookLM-style backend where documents are uploaded per project and answered against. The ingestion pipeline (upload → store → load → chunk → persist) is complete; embedding and vector retrieval are in active development.
- Designed a layered FastAPI service (routes → controllers → models) around a single error boundary: typed domain exceptions carry their own HTTP status and one handler translates them, leaving every route free of `try/except`.
- Implemented async ingestion on MongoDB with Motor across three indexed collections — uploads validated on MIME type and size, then chunked with LangChain's `RecursiveCharacterTextSplitter` and written in unordered 100-document batches.
- Added production-style observability: ASGI middleware stamping a correlation id on every request and every log line it produces, plus configurable text/JSON logging with rotating file output.

Keywords: RAG, FastAPI, MongoDB, Motor (async), LangChain, Pydantic, Docker Compose, REST API Design, Structured Logging, Layered Architecture.

Volleyball Group Activity Recognition

Progressively complex baselines on the Volleyball Dataset (55 videos, 4,830 clips, 60+GB), each isolating one component; reported as test accuracy / macro-F1:

- **B1** — single middle frame, fine-tuned ResNet-50: **62.6%** / 0.63.

- **B3** — person crops, frozen backbone, concat-pool + MLP: **60.7%** / 0.59.
- **B4** — 9-frame clip, frozen backbone + LSTM over time: **66.1%** / 0.67.
- **B5** — per-player LSTM, pooled player summaries + MLP: **66.3%** / 0.62.
- **B6** — players pooled per frame, scene LSTM + skip Conv1d: **70.5%** / 0.69.
- **B7** — hierarchical player-LSTM and scene-LSTM with skip connections: **73.8%** / 0.70.
- **B8** — B7 plus team-split pooling: **85.6%** / 0.86 (best; exceeds the CVPR 2016 result, 81.9%).

Keywords: Group Activity Recognition, Volleyball Dataset, PyTorch, ResNet-50, LSTM, Hydra, LMDB, TensorBoard, DataParallel, AdamW, Transfer Learning, Computer Vision.

Credit Card Fraud Detection on Severely Imbalanced Data

- Tackled extreme class imbalance ($< 0.2\%$ fraud rate) by benchmarking 11 ML models, prioritizing algorithmic solutions over data resampling, using the popular Kaggle Credit Card Fraud dataset with 285K transactions.
- Implemented custom Focal Loss function in PyTorch from mathematical foundations, training both Logistic Regression and Neural Network models to specifically address minority class detection without data-level interventions.
- Conducted systematic model comparison across diverse algorithm families including ensemble methods (Random Forest), boosting algorithms (XGBoost, LightGBM, CatBoost, AdaBoost, Gradient Boosting), and deep learning approaches.
- Achieved optimal fraud detection with XGBoost (F1: 0.89, AP: 0.87, ROC-AUC: 0.98), outperforming baseline Logistic Regression by 16% in F1-score while maintaining high precision on the minority fraud class.

Result: XGBoost reaches **F1 0.89, ROC-AUC 0.98** on the minority fraud class — a 16% F1 gain over the Logistic Regression baseline.

Keywords: Imbalanced Classification, Focal Loss Implementation, XGBoost, PyTorch, Model Benchmarking, Fraud Detection, Ensemble Methods.

New York City Taxi Trip Duration Prediction

- Engineered comprehensive feature sets by integrating 5 external data sources including NYC traffic volumes (2016 DOT data), historical weather patterns, geospatial borough boundaries, and holiday calendars, transforming 10 original features into 55 engineered features.
- Deployed a local OSRM routing server using Docker to query OpenStreetMap data, generating real-world routing features (shortest path distance and estimated travel time) for 1M+ trip records.
- Implemented a complete preprocessing pipeline featuring log transformation of target variable to reduce variance, strategic outlier removal (trips $> 1000\text{km}$), one-hot encoding, and polynomial feature generation (degree 2) to capture non-linear interactions.
- Achieved strong predictive performance with Ridge Regression, securing RMSE of 0.397 and R^2 score of 71.6% on test set (224,734 samples), demonstrating effective feature engineering on large-scale geospatial data.

Result: Ridge Regression achieves **RMSE 0.397, R^2 71.6%** on 224,734 held-out trips, driven by feature engineering from five external sources.

Keywords: Feature Engineering, OSRM API, Docker, Geospatial Analysis, External Data Integration, Ridge Regression, Time Series Features.

EDUCATION

Bachelor of Science, Mechatronics Engineering
Helwan University

2019 – 2024

- Graduated with Honors (Grade: Very Good, GPA: 3.0/4.0).
- Engineered a [mathematical model](#), implemented it on MATLAB for a green hydrogen generator to control input voltage and current while predictively monitoring critical safety parameters (e.g., operating temperature).